6.2 Prime Factorization

Lesson Introduction

Benchmark	MA.6.NSO.3.4 Express composite whole numbers as a product of prime factors with natural number exponents.		Learning Target	• I can express composite whole numbers as a product of prime factors.
Benchmark Coherence	Building On		Working Toward	
	MA.4.AR.3.1 Determine factor pairs for a whole number from 0 to 144. Determine whether a whole number from 0 to 144 is prime, composite, or neither.		N/A	
Essential Questions	• What is a factor? • What is a prime factor? • How do you determine the prime factorization of a composite whole number?			
Lesson Narrative	In this lesson, students use their prior knowledge of factors and prime numbers to explore finding the prime factors of composite numbers using factor trees. Students extend their understanding of prime factorizations by expressing composite whole numbers as the product of prime factors using exponents.			
Component Summary	6.2.1 Warm-Up (~5 min.)	Select which scenario you prefer and support the choice mathematically *(SE p. 238)*	6.2.4 Guided Practice (10 min.)	Practice expressing composite numbers as a product of prime factors *(SE p. 241)*
	6.2.2 Collaboration (5-10 min.)	Represent composite numbers as products of factors and explain how the factor pairs relate *(SE p. 239)*	6.2.5 Your Turn! (10 min.)	Independent practice expressing three-digit composite numbers as a product of prime factors *(SE p. 241)*
	6.2.3 Guided Instruction (10 min.)	Express composite numbers as a product of prime factors *(SE p. 239)*	6.2.6 Wrap-Up (~5 min.)	Demonstrate understanding of writing the prime factorization of composite numbers *(SE p. 242)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Prime number • Prime factor • Prime factorization <u>Additional Terms:</u> N/A	• (Optional) Factor tree graphic organizers	• If differentiating for varying writing abilities, consider preparing sentence starters/stems. • Consider prepping a list of prime factors for students to reference as they build fluency.

Teacher Tips

Common Misconceptions	Things to Consider
• Students may struggle to see how the factor pairs relate to each other or may struggle to explain the connection in writing in the 6.2.2 Collaboration. • Students without a strong grasp of multiplication may struggle to find the prime factors of three- and four-digit numbers.	• Consider reviewing the definition of prime number before the lesson to activate prior knowledge. • Consider providing factor tree graphic organizers, particularly in 6.2.3 Guided Instruction, 6.2.4 Guided Practice, 6.2.5 Your Turn!, and 6.2.6 Wrap-Up. • Consider providing a list of prime factors for students to reference throughout the lesson as an additional scaffold.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Think-Pair-Share In 6.2.4 Guided Practice and 6.2.5 Your Turn!, have students first complete problems independently and then discuss their work with a partner. These discussions will build student confidence and understanding.	**MA.K12.MTR.5.1: Patterns and Structures** In 6.2.2 Collaboration, the task lends itself well to students using patterns and structure to help understand and connect mathematical concepts in building toward understanding of prime factorization. **MA.K12.MTR.4.1: Discussion and Reflection** In question 2 of 6.2.4 Guided Practice, students have an opportunity to analyze the errors in a sample of student work. Students identify the error, write a letter to the student explaining the error, and then correct the error.

Differentiate Instruction

Consider providing students with factor tree graphic organizers to provide a visual entry point to help them express composite whole numbers as products of prime factors. Discussions during 6.2.2 Collaboration provide auditory entry points.

Consider providing a choice of composite numbers for 6.2.5 Your Turn! question 3 for students who may struggle to come up with a number on their own. This will allow students to focus on the process of writing the prime factorization.

Lesson Notes:

6.2.1 Warm-Up: Would You Rather?

Component Overview: *Students select which of two scenarios they prefer and support their choice mathematically.*

6.2.1 Warm-Up: Would you Rather?

1. Which of the following would you rather choose? Justify your reasoning using mathematics.

Deposit $\$3$ in the bank and have it triple each week for 4 weeks.	Deposit $\$4$ in the bank and have it quadruple each week for 3 weeks.

Answers will vary.
If you deposit $\$3$ in the bank and have it triple every 4 weeks you will have the following: Week 1 $(\$3)$. Week 2 $(3 \cdot 3 = \$9)$. Week 3 $(9 \cdot 3 = \$27)$. Week 4 $(27 \cdot 3 = \$81)$.
If you deposit $\$4$ in the bank and have it quadruple each week for 3 weeks you will have the following: Week 1 $(\$4)$, Week 2 $(4 \cdot 4 = \$16)$. Week 3 $(16 \cdot 4) = \$64$.

I would rather have the $3 tripled for 4 weeks because I would end up with more money in the end.

Consider asking students the following:

- *How can you represent each scenario using multiplication?*
- *How much money will be in the bank after 4 weeks?*
- *How much will be in the bank after 3 weeks?*

6.2.2 Collaboration: Exploring Factors

Component Overview: *Students work with a partner to write whole numbers as factor pairs. Students then draw connections among the factor pairs.*

6.2.2 Collaboration: Exploring Factors

Work with your partner on the following.

1. Write the number 24 as a factor pair that uses the greatest factor other than itself.
   $2 \cdot 12$

2. Write the number 12 as a factor pair that uses the greatest factor other than itself.
   $2 \cdot 6$

3. Write the number 6 as a factor pair that uses the greatest factor other than itself.
   $2 \cdot 3$

4. Write the number 3 as a factor pair that uses the greatest factor other than itself.
   There is no way to do this because the only factors of 3 are 3 and 1.

5. How did each of the above tasks, problems 2-4, relate to the factor pair that came before it?
   Answers will vary. With the exception of task number 4, each factor pair has 2 as a factor. Each factor pair are factors of 24. The greatest factor of each task is also a factor of the greatest factor of the pair of the previous task.

6. What was different about rewriting the number 3 as a factor pair using the greatest factor other than itself?
   Answers will vary. The number 3's greatest factor other than itself is 1.

Consider your students when planning this Collaboration to ensure equitable opportunity for contribution. Students may either just work independently side by side instead of collaborating, or one student may dominate the conversation. Prompt discussions during this component using intentional pauses, or give students specified roles and questions to then compare with their partner or a small group.

MA.K12.MTR.5.1: Patterns and Structure
Encourage students to make use of structure in making meaningful connections for question 6.

6.2.3 Guided Instruction: Prime Factorization

Component Overview: *Students review the definition of prime numbers and learn the definitions of prime factorization and prime factors. Students use factor trees to determine the prime factorization of composite numbers, representing repeated multiplication in the prime factorizations using exponents.*

6.2.3 Guided Instruction: Prime Factorization

Each of the factor pairs from the Collaboration can be used to rewrite 24 using more than one pair of factors.

For example, the first factor pair was $2 \cdot 12$, so 24 can be rewritten as $24 = 2 \cdot 12$.

Because 12 can be written as $2 \cdot 6$, 24 can also be rewritten as $24 = 2 \cdot 2 \cdot 6$.

1. Since 6 can be expressed as $2 \cdot 3$, then 24 can be

   rewritten as $24 = 2 \cdot 2 \cdot 2 \cdot 3$.

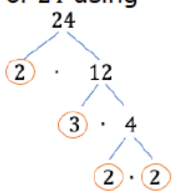
This is called the ***prime factorization*** of 24 because all of the factors are ***prime numbers***.

A ***prime number*** is a whole number greater than 1 that is not divisible by any whole number other than 1 and itself.

*A **prime factor** is a factor that is a prime number.*

Prime factorization is the expression of a number as the product of prime factors.

2. Cory determined the prime factorization of 24 using different factor pairs.



   a. Explain whether Cory's method changed the final result.
      Cory's method would not change The result. The prime factorization remains: $2 \cdot 2 \cdot 2 \cdot 3$.
   b. Why did Cory circle the values he did?
      Cory is identifying the prime factors.

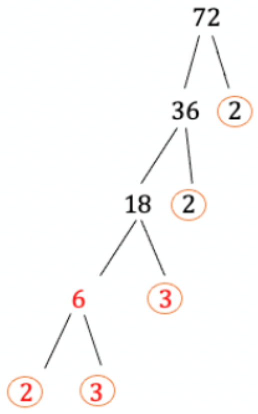
Utilize the Study Expert-led Glossary Videos for additional support with these glossary terms in multiple languages.

How do you determine if a number is prime?
If a prime factorization has repeated prime factors, how do you represent them?

Consider providing a list of prime factors for students to reference as they begin the process of prime factorization of whole numbers as additional scaffolded support.

6.2.3 Guided Instruction: Prime Factorization (continued)

3. Find the prime factorization of 72 by completing the work below.



The prime factorization of 72 is $2 \cdot 2 \cdot 2 \cdot 3 \cdot 3$.

4. Consider the expression $2 \times 2 \times 2 \times 3 \times 3 \times 5$.

   a. What is the product of $2 \times 2 \times 2 \times 3 \times 3 \times 5$?
   360

   b. The prime factorization of 360 is $2 \times 2 \times 2 \times 3 \times 3 \times 5$.

Numbers with multiple repeating prime factors can be written using exponents. For example, the prime factorization of 360 can be written as $2^3 \times 3^2 \times 5$.

5. Why is it helpful to write the prime factorization with exponents?
   Answers will vary. When there are many of the same factor(s), the prime factorization becomes very long to write. Prime factorization with exponents allows for the factors to be seen more easily.

For students who may struggle with multiplication, encourage them to start by multiplying two factors at a time, chunking the expression into more manageable parts.

Consider using the Think-Pair-Share routine before question 4 to provide students an opportunity to discuss the reasons it is helpful to use exponents when writing prime factorization.

6.2.4 Guided Practice: Prime Factorization

Component Overview: *Students find the prime factorization of composite numbers and represent them using exponents. Students also analyze the errors in a sample of student work.*

6.2.4 Guided Practice: Prime Factorization

1. Find the prime factorization of each value. Use exponents where appropriate.

   a. 132 — $2^2 \cdot 3 \cdot 11$
   b. 270 — $2 \cdot 3^3 \cdot 5$
   c. 882 — $2 \cdot 3^2 \cdot 7^2$

2. Sophie said the prime factorization of 210 is $3 \cdot 7 \cdot 10$.

   a. Determine the error Sophie made. Then, write her a brief note explaining her error and how she can learn from it.

   Means
   I
   Start
   To
   Acquire
   Knowledge
   Experience
   Skills

   Answers will vary. Sophie, you are almost there! When factoring a number, each factor must be a prime number. 10 is not a prime number. Recall that a prime number is a whole number greater than 1 that is not divisible by any whole number other than 1 and itself. To complete your steps, 10 will need to be factored using 2 and 5, which are prime numbers.

   When factoring, remember to ask yourself, "is each factor a prime number?" You got this!

   b. What is the prime factorization of 210?
   $2 \cdot 3 \cdot 5 \cdot 7$

Consider providing sentence starters on question 2 for students who struggle writing an explanation of the error.

- *"When finding the prime factors, you should…"*
- *"The ____ is wrong because…"*
- *"Next time, it might be helpful if you…"*

Have a few students share the error they identified and how to correct it.

Encourage students to finish their note with encouragement for Sophie to foster growth mindset.

6.2.5 Your Turn!

Component Overview: *Students independently practice writing the prime factorization of composite whole numbers.*

6.2.5 Your Turn!

1. Find the prime factorization of each value. Use exponents where appropriate.

   a. 904 — $2^3 \cdot 113$

   b. $1{,}125$ — $3^2 \cdot 5^3$

   c. 80 — $5 \cdot 2^4$

2. Which number includes the largest exponent when written in prime factorization? Explain your answer.

5,670	64
$2 \cdot 3^4 \cdot 5 \cdot 7$	2^6

   Written in prime factorization, 64 *has the largest exponent,* 6*. There are* 6 *factors of* 2 *in* 64*. There are* 4 *factors of* 3 *in the prime factorization of* $5{,}670$*.*

3. Find a number that meets the following conditions.
   - The value of the number is less than 150.
   - One of its prime factors has an exponent of 3.
   - When expressed in exponential form, its prime factorization has only two unique bases.

   Answers will vary. $2^3 \cdot 5 = 40$, $3^3 \cdot 2^2 = 108$

If time allows after completing the Your Turn!, have students compare their answers to question 2 with a partner. Elicit any responses of students who changed their answer because of something their partner shared.

If students finish early, have them come up with additional examples that meet the requirements in question 3 and challenge a family member at home or another teacher with their number.

6.2.6 Wrap-Up: Ticket Out the Door

Component Overview: *Students demonstrate their understanding of the lesson learning targets. Use data from this formative assessment to determine which students need support on this benchmark.*

6.2.6 Wrap-Up: Ticket Out the Door

1. Find the prime factorization of each value. Use exponents where appropriate.

   a. 168
   $2^3 \cdot 3 \cdot 7$

   b. 1,040
   $2^4 \cdot 5 \cdot 13$

   c. 144
   $2^4 \cdot 3^2$

Consider providing factor tree graphic organizers for students to use to show their work for more information when assessing their progress.

If students need additional scaffolding, consider having them make two attempts. In their first attempt, they should show their work and mark as "without a calculator." With their second attempt, they should show work marked as "with a calculator."